Mr. Qihua Dong (His/He/Him)

Personal page: https://dddraxxx.github.io Email: dongqh078[at]gmail.com

Github: https://github.com/dddraxxx

LinkedIn: https://www.linkedin.com/in/qdong30/

Education

Northeastern University, Boston, USA Sep 2023 - Now

- **PhD** in **Computer Engineering** under the supervision of Prof. Raymond Fu
- Interested topics: Multi-modal LLM, Computer Vision, Artificial Intelligence
- GPA: 3.96/4.00

City University of Hong Kong, Hong Kong SAR, China Sep 2017 - Jul 2021

- BSc in Computer Science with Mathematics Minor; overall GPA 3.89/4.3 (Top 10% official proofed)
- Featured courses: Mathematical Analysis; Enhanced Calculus and Linear Algebra; Discrete Mathematics; Statistic Processes; Data Structures and Advanced Programming; Design and Analysis of Algorithms

Publications

- **Dong Qihua**, Kuo Yang, Lin Ju, Handong Zhao, Yitian Zhang, Yizhou Wang, Huimin Zeng, Jianglin Lu, Yun Fu. "Ref-Adv: Exploring MLLM Visual Reasoning in Referring Expression Tasks". On submission, *2025*.
- **Dong Qihua**, Luis Figueroa, Handong Zhao, Kushal Kafle, Jason Kuen, Zhihong Ding, Scott Cohen, Yun Fu. "CoT Referring: Chain-of-Thought Grounding Improve Localization in Referring Expression Tasks". On submission, *2025*.
- Mingyuan Zhang, Yue Bai, Huan Wang, Yizhou Wang, **Qihua Dong**, Yun Fu. "Boosting Large Language Models with Mask Fine-Tuning". https://arxiv.org/abs/2503.22764 , *2025*.
- Du Hao, **Dong Qihua**, Xu Yan, Liao Jing. "Contrastive Feature Extraction and Shared-Unique Clustered Attention for Multimodal 3D Medical Image Analysis". Submitted to ***Journal of Biomedical and Health Informatics****, 2024.*
- **Dong Qihua**, Du Hao, Song Ying, Xu Yan, Liao Jing. "Preserving Tumor Volumes for Unsupervised Medical Image Registration". ***Proc. ICCV****, 2023*.
  Paper link: https://arxiv.org/pdf/2309.10153.pdf
- He Ruozhen*, **Dong Qihua***, Lin Jiayin, Rynson Lau. "Weakly-Supervised Camouflaged Object Detection with Scribble Annotations". ***Proc. AAAI****, 2023*.
  Paper link: https://arxiv.org/pdf/2207.14083v2.pdf
- Du Hao*, **Dong Qihua*,** Xu Yan, Liao Jing. "Weakly-Supervised 3D Medical Image Segmentation using Geometric Prior and Contrastive Similarity". ***IEEE Transactions on Medical Imaging****, 2023*.
  Paper link: https://arxiv.org/pdf/2302.02125.pdf

*: The authors with * contributed equally to the work

Intern Experience

Amazon AGI Foundation Team, WA, USA May 2025 – Now

Mentor: Hao Yang (https://sites.google.com/site/lancelot365/)

- Position: Applied Scientist Intern
- Brief: Research on **visual agents** with **tool use** and **reasoning**, using **Reinforcement Learning** and **Multimodal LLM**.
- Responsibilities: Building the visual agents that can use **multiple tools** (zoom-in, draw-box, rotate) and conduct visual reasoning with **GRPO**. Build the cold-start and reinforcement data from scratch, train on 7B and 32B models, and improve model performance across **a wide range of general benchmarks** (chart, doc, high-resolution and son on). Novel reward design and masking is applied to enable stable training.

Adobe Research, CA, USA May 2024 – Feb 2025
Mentor: Luis Figueroa (https://luisf.me/)

- Position: Research Intern
- Breif: Research on referring image segmentation with Multimodal LLM.
- Responsibilities: Developing an advanced multi-modal large language model (MLLM) for image segmentation tasks, focusing on segmentation on **complex referring expression**. Proposed a structured, **chain-of-thought strategy** enhancing model reasoning across modalities. The approach incorporates new annotations for existing datasets and a novel benchmark tailored for complex referring cases.

Research Experience

Multi-modal LLM in **visual understanding** and **reasoning** Oct 2023 – Now

- Supervisor: Prof. Raymond Fu, Northeastern Univ.
- Research brief: Developed new multi-modal large language model (LLM) capable of visual grounding and reasoning, combined with spatial cognition. The final goal is building an intelligent agent capable of solving real-world problems with no costly finetuning but just some in-context examples. Visual understanding and thinking is critical, since it prevents hallucination and allow zero-shot generalization in complex tasks.

Study on 3D Medical Image May 2022 – Mar 2023

- Supervisor: Prof **Jing Liao**, Prof **Yan Xu**, City Univ. of HK
- Research 1: Preserving Tumor Volume in 3D Medical Image Registration
    - Proposed a novel registration neural network that preserves the volume of tumors without the loss of alignment accuracy, allowing registration network to have wider clinical application. A new volume-preserving loss is designed specifically for the task.
- Research 2: Dynamic Transformer for 3D Medical Image Segmentation
    - Research brief: Designed an efficient and accurate transformer that adapts to medical image segmentation with dynamic token generation, in which only foreground-related tokens are further split and attended. Our method improves SOTA Dice scores of transformers on two popular datasets in medical images.
- Research 3: Weakly-Supervised 3D Medical Image Segmentation Framework
    - Research brief: Designed a new weakly supervised framework to segment medical images with bounding-box labels. The framework exploited the geometric prior of organs by using template organs in point-cloud form, which provides delicate shape guidance for segments. Besides, pretraining of the feature head was performed to learn contrasted features between objects and background. The work improves SOTA performance by 12.5% on Dice on two popular datasets in medical images.

Weakly-Supervised Camouflaged Object Detection Dec 2021 - Jun 2022

- Supervisor: Prof Rynson Lau, City Univ. of HK
- Research brief: Proposed the first weakly-supervised task and dataset in camouflaged object detection. It contains the novel consistency loss that corrects an ignored bias in previous works. The feature loss leveraging semantic features to infer concealed objects. Our model outperforms SOTA methods on COD benchmarks with an average improvement of 11.0% on MAE.

Undergrad Work Experience

Texas A&M University, TX, USA Oct 2022 – Jan 2023

- Supervisor: Prof Wenping Wang (https://engineering.tamu.edu/cse/profiles/Wang-Wenping.html)
- Position: Research Assistant (Remote)
- Responsibilities: Conducting research related to faces in computer vision and computer graphics.

City University of Hong Kong, Hong Kong SAR, China Aug 2021 – Aug 2023

- Supervisor: Prof Jing Liao (https://www.cityu.edu.hk/stfprofile/jingliao.htm)

- Position: Research Assistant (Full-time)
- Responsibilities: Engaging the research projects in computer vision, the primary research domain is 3D Medical Image Analysis and Weakly-Supervised Learning.

City University of Hong Kong, Hong Kong SAR, China Sep 2020- Feb 2021

- Supervisor: Prof Jianping Wang (https://www.cs.cityu.edu.hk/~jianwang/)
- Position: Student Research Assistant (Part-time)
- Responsibilities: Research works in computer science: the major research topics are related to system scheduling schemes for self-driving tasks.

Sik Sik Yuen, Hong Kong SAR, China Sep 2019- Jun 2020

- Position: Full-stack Programmer (Intern)
- Responsibilities: Mobile phone application development project including AI functions and web service.

Honours and Scholarship

2021	**First class honor**, Bachelor's degree of City University of Hong Kong, HK
2019	**Bronze Medal**, The 2019 **ICPC** Asia Nanchang Regional Contest, Jiangxi Normal University, China
2017 - 2021	**Dean's List**, City University of Hong Kong, HK
2017 - 2021	**Full Tuition Scholarship**, City University of Hong Kong, HK

Skills

- Computer languages Python, C++, Matlab, R, Javascript, Java, CSS
- Tools and library PyTorch, Numpy, Detectron2, Skimage, mmDet, Monai, Scikit-learn,
- English skills TOEFL 110 (Nov 2022); GRE 330 (+3.5)